# Essays

|!|BCEEC0150000001|!|
Describe the development of root from a seed. (11 marks)
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| Content | Description | Marks |
|---|---|---|
| Seed germination | - Cells divide at the apical meristem of the radicle by mitotic cell division. (1)<br>- Water activates the enzymes in the cotyledons. (1)<br>- Stored starch in the cotyledons is broken down into sugars due to the catalysis by the enzymes. (1)<br>- The sugars are broken down in respiration to provide energy for the synthesis of substances and cell division in the radicle. (1)<br>- The sugars are also used to form new cellulose cell walls of the cells found in the apical meristem of the radicle. (1)<br>- Stored proteins are broken down into amino acids, which are used to make new protein molecules for cell membranes and cytoplasm of the cells found in the apical meristem of the radicle. (1)<br>- Radicle lengthens as a result of cell division. (1) | (Max. 4) |
| Growth and development of the root tip | - Cells elongate at the region of elongation by drawing in water by osmosis. (1)<br>- Small vacuoles containing water in each cell fuse to form a large vacuole as more water is absorbed. (1)<br>- As the cells elongate, there is a gradual thickening of the cellulose cell wall. (1)<br>- The cells may then differentiate into specialized tissues (e.g. xylem, phloem and the epidermis with root hairs) for different functions. (1)<br>- After mitotic cell division carried out by the vascular cambium, secondary growth results, in which the thickness of the root increases. (1) | (Max. 4) |
| **Communication** | | (Max. 3) |
| | | (11 marks) |

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| Skill assessed | Marks |
|---|---|
| Understanding basic concepts | 8 |
| Communication | 3 |

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